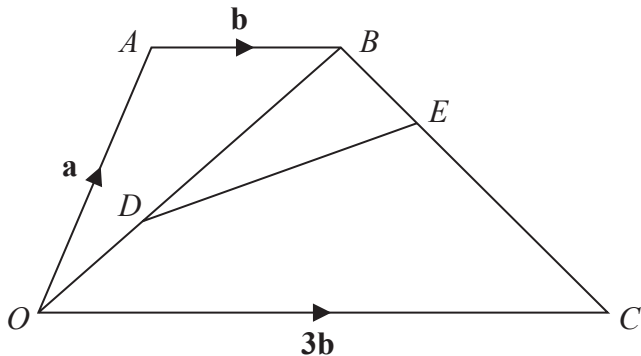


18 $OABC$ is a trapezium.



$$\vec{OA} = \mathbf{a}$$

$$\vec{AB} = \mathbf{b}$$

$$\vec{OC} = 3\mathbf{b}$$

D is the point on OB such that $OD:DB = 2:3$

E is the point on BC such that $BE:EC = 1:4$

Work out the vector $\vec{DE}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Give your answer in its simplest form.

(Total for Question 18 is 4 marks)